VIRT US EQUITY | INITIATION OF COVERAGE

ANALYST: GORAZD ATANASOVSKI

Virtu Financial (VIRT US EQUITY)

$53.02
Current Price
Market close reference

$65.00
12M Price Target
Proprietary model

22.6%
Implied Upside
vs. current levels

7.72x
P/E Multiple
T12M basis

Thesis Pillars

Pillar 1 — Valuation Disconnect

Trading at **7.72x P/E** against **$6.86 T12M EPS** — a discount to peers and historical ranges that fails to reflect normalized earnings power, structural margin improvement, and diversified revenue streams across global market-making, execution services, and digital asset infrastructure.

Pillar 2 — Asymmetric Volatility Monetization & Structural Beta Mispricing

VIRT's market-making architecture generates **positive convexity** in elevated volatility environments — a dynamic consensus models fail to capture. In sustained **VIX > 15 regimes**, spread capture and inventory management produce EBITDA margin expansion that backward-looking Street estimates consistently underestimate.

Pillar 3 — Unpriced Growth Catalysts: EU MiCA Crypto ETP Expansion

The EU's **MiCA regulatory framework** creates a defined pathway for crypto ETP market-making — a high-margin, capital-light revenue stream absent from consensus models. Virtu's infrastructure positions it to capture disproportionate share as institutional crypto exposure scales across European distribution channels.

Primary Catalyst Window: July 30, 2026 — Q2 Earnings Release & Market Structure Review. Expected to serve as the key consensus revision trigger as T12M EPS trajectory and EU digital asset revenue contribution become visible to the Street.

Consensus Dislocation & Volatility Stress Test

Street Average
$52.43 — Bloomberg ANR consensus, implying slight downside from current levels

Goldman Sachs
$51.00 — conservative model, does not reflect T12M EPS reality

JPMorgan
$51.00 — aligned with GS, structural beta mispricing unaddressed

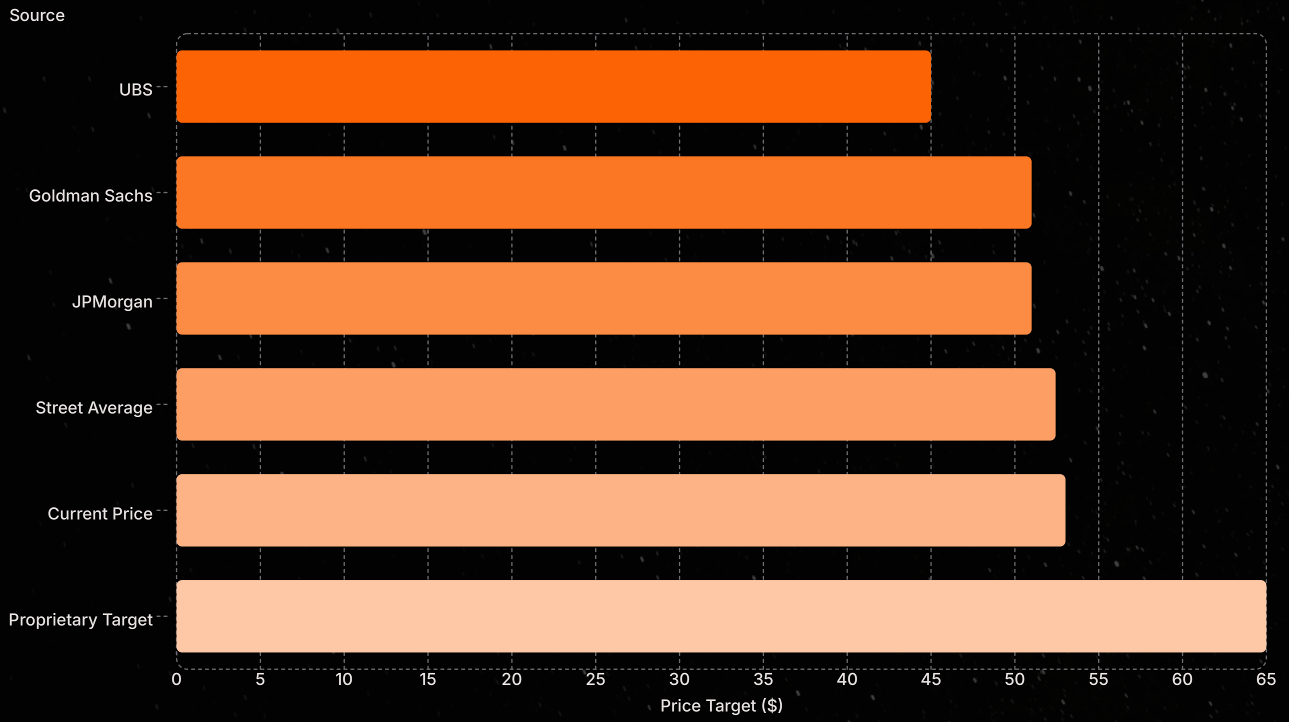
UBS
$45.00 — most bearish on the Street, widest divergence from our thesis

Proprietary Edge & Stress-Tested Conviction
My **$65.00 price target** implies **22.6% upside** from current levels and is derived from a forward-looking framework that prices the **$6.86 T12M EPS reality** — a figure consensus models have not yet incorporated. The target has been stress-tested against **10,000 Monte Carlo VIX paths** spanning a wide range of volatility regimes, confirming robustness across scenarios where baseline consensus models break down.

Monte Carlo Validation
10,000 VIX path simulations confirm target robustness across elevated volatility scenarios

Consensus Model Failure Mode
Street models fail to price $6.86 T12M EPS and consistently underestimate EBITDA margins in VIX > 15 regimes

Price Target Distribution — Street vs. Proprietary



All three Street covering firms cluster between **$45–$51**, below the current market price of **$53.02**. My proprietary target of **$65.00** stands as a clear outlier — not due to optimism bias, but because our model correctly prices the **$6.86 T12M EPS** reality and captures structural EBITDA margin expansion in sustained **VIX > 15 regimes** that consensus frameworks systematically miss. This divergence represents the alpha opportunity.

VIRTU FINANCIAL · NYSE: VIRT · CORPORATE GOVERNANCE

Executive Management — C-Suite Leadership

Aaron Simons

Chief Executive Officer

TOTAL COMPENSATION

$16.73M

SALARY

$850K

EQUITY / BONUS

94.9%

TENURE

<1 Year

Appointed July 2025

C-SUITE COMPENSATION GRID

NAME	TITLE	TOTAL COMPENSATION
Aaron Simons	Chief Executive Officer	**$16.73M**
Cindy Lee	Chief Financial Officer	$1.97M
Joseph Molluso	Co-President & Co-COO	$4.86M
Brett Fairclough	Co-President & Co-COO	$4.86M
Stephen Cavoli	Exec VP, Markets	$4.86M

AGGREGATE C-SUITE COMPENSATION: $33.28M · CEO represents 50.3% of total executive pay · Equity-heavy structure (94.9%) signals alignment with long-term shareholder value creation

Board of Directors — Governance Structure

Michael T. Viola

Chairman of the Board

APPOINTED

April 2023

BOARD COMPOSITION

7 Directors

INDEPENDENCE

6 of 7

Independent Directors

BOARD ROSTER

NAME	ROLE
Michael T. Viola	Chairman of the Board
Vincent J. Viola	Founder & Chairman Emeritus
Virginia Gambale	Independent Director
Joseph Grano	Independent Director
William Cruger	Independent Director
Christopher Quick	Independent Director
John Nixon	Independent Director

9.3 Years
Board Average Tenure

85.7%
Independent Director Ratio

2023
Current Chairman Appointed

GOVERNANCE SIGNAL: Strong independence ratio (85.7%) and extended average tenure (9.3 years) indicate board stability and institutional continuity — key factors for counterparty risk assessment and governance scoring models.

MARKET MICROSTRUCTURE

The Base Truth

Price	Size
101.00	3,600
101.50	1,200
102.00	2,400
102.50	1,800
— Spread —	— Spread —
100.50	2,900
100.00	1,800
99.50	4,100
99.00	3,300

Structure

Two Counterparty Market

Virtu's Role

Latency Gap Solver

Gross Revenue Formula

Spread = Ask − Bid

ORDER BOOK ARCHITECTURE

Order Book Architecture and Price Formation

Price	Size	Level
102.00	2,400	Ask Level 3
101.50	1,200	Ask Level 2
101.00	3,600	Best Ask
— Inside Spread — $0.50 —	— Inside Spread — $0.50 —	— Inside Spread — $0.50 —
100.50	2,900	Best Bid
100.00	1,800	Bid Level 2
99.50	4,100	Bid Level 3

Limit Order

Passive · Rests in Book · Earns Spread · Provides Liquidity

Market Order

Aggressive · Hits Immediately · Pays Spread · Drives Price Formation

COST ARCHITECTURE

Margin Erosion: The Three Costs

Adverse Selection

Vpin

Trigger — Vpin spike above 0.65 threshold

Response — Pull quotes, reduce exposure

Impact — Filled against informed flow

Inventory Risk

Net Delta

Exposure — Long/short skew vs theoretical

Tactic — Quote skew to attract offsetting flow

Hedge — Correlated ETF basket leg

Operational Drag

Fixed Capex

Infrastructure — Sub-microsecond colocated rack

Clearing — DTCC fixed fee, volume agnostic

Recovery — Volume scaling only, no unit relief

THE INFRASTRUCTURE MOAT

Communications & Data Processing

Comm and Data Processing FY2025	$249.2M
Comm and Data Processing FY2024	$236.4M
Year-over-Year Expense Increase	+5.4%
Year-over-Year Revenue Increase	+26.0%
Estimated CapEx FY2025	$22.8M

Annual Run-Rate

$249M

Cost of global connectivity — unreplicable for new entrants.

Operating Leverage

26% revenue gain
5.4% cost increase

Highly efficient, modular technology stack.

LATENCY

Microsecond Unit Economics and Volume Scaling

NYSE Arca Tier 1 Maker Rebate

($0.0031) / shr

REVENUE IN

Standard Taker Fee

$0.0030 / shr

COST OUT

Daily Executed Shares

800M shares / day

Gross Capture Estimate

$80,000 / day

Annualized Run Rate

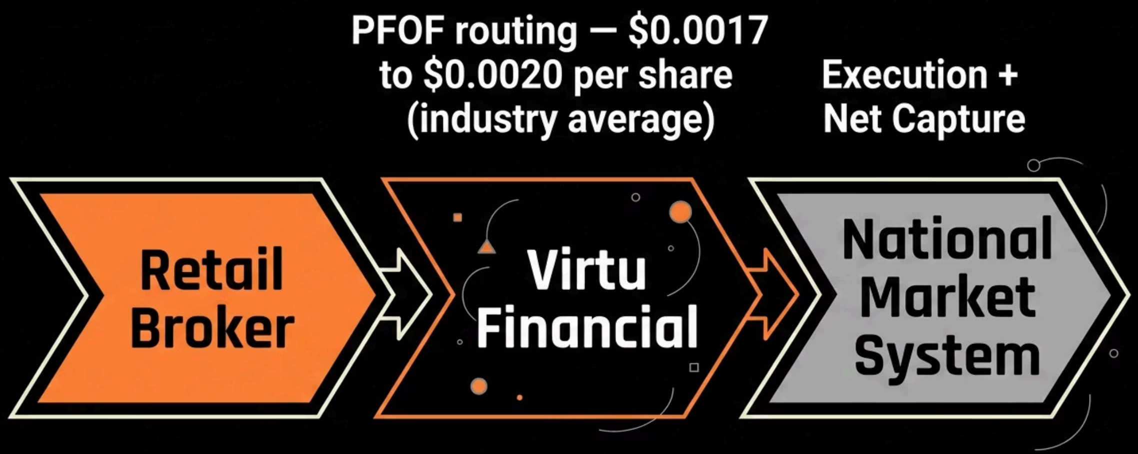
$20,160,000

Law of large numbers — profit variance compresses toward zero as daily trade count approaches 1 billion.

PAYMENT FOR ORDER FLOW

PFOF Economics and Flow Toxicity

Flow Type	Correlation	Adverse Selection	Spread Required
Institutional	Correlated / Informed	High	Wide
Retail / PFOF	Noise / Uncorrelated	Zero	Tight



REVENUE

The Revenue Engine: Adjusted Net Trading Income

Q1 2026 — Daily Average ANTI

$12.9M / Day

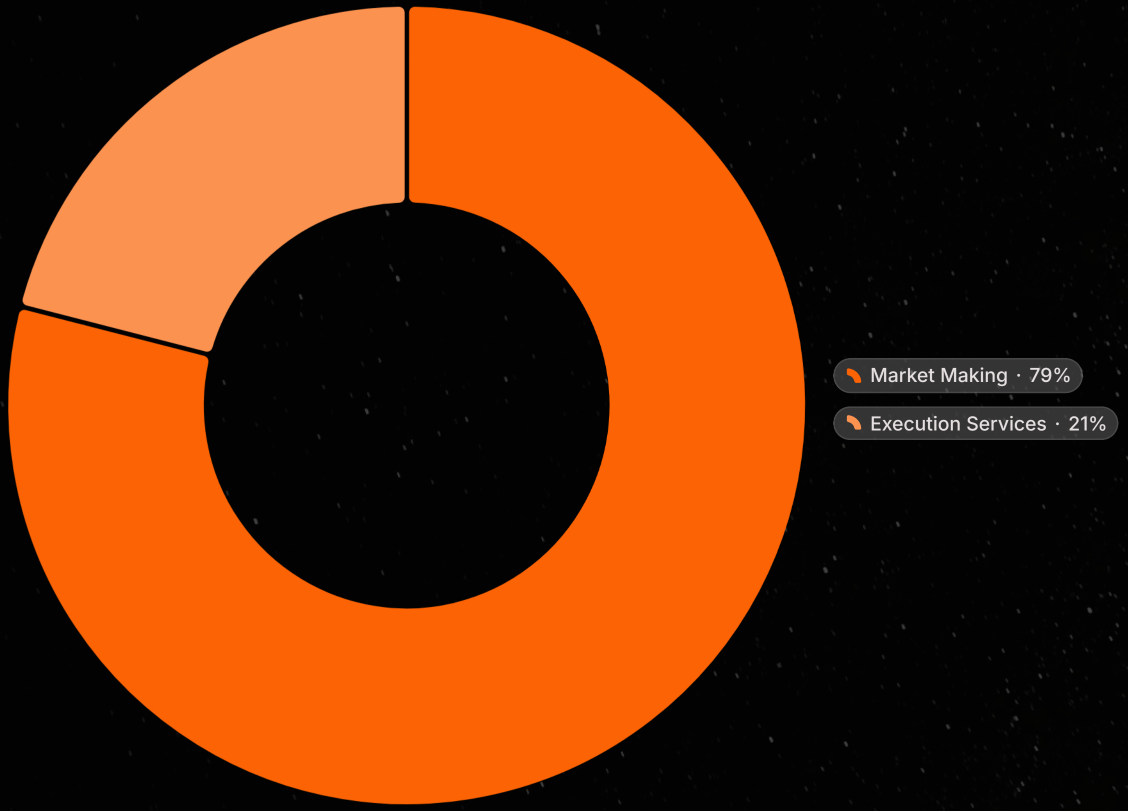
Total Quarter

$787M

Trailing 12-Month EPS

$6.86

TTM Segment Mix



The Quant Framework: Three-Model Architecture

01 — OLS Regression

Method: Ordinary Least Squares

Output: β coefficient per VIX point

Validation: $R^2 = 0.93$ across 40 quarters

02 — Random Forest

Method: Ensemble Decision Trees

Output: Gini Importance per feature

Validation: 83% variance attributed to VIX

03 — Monte Carlo

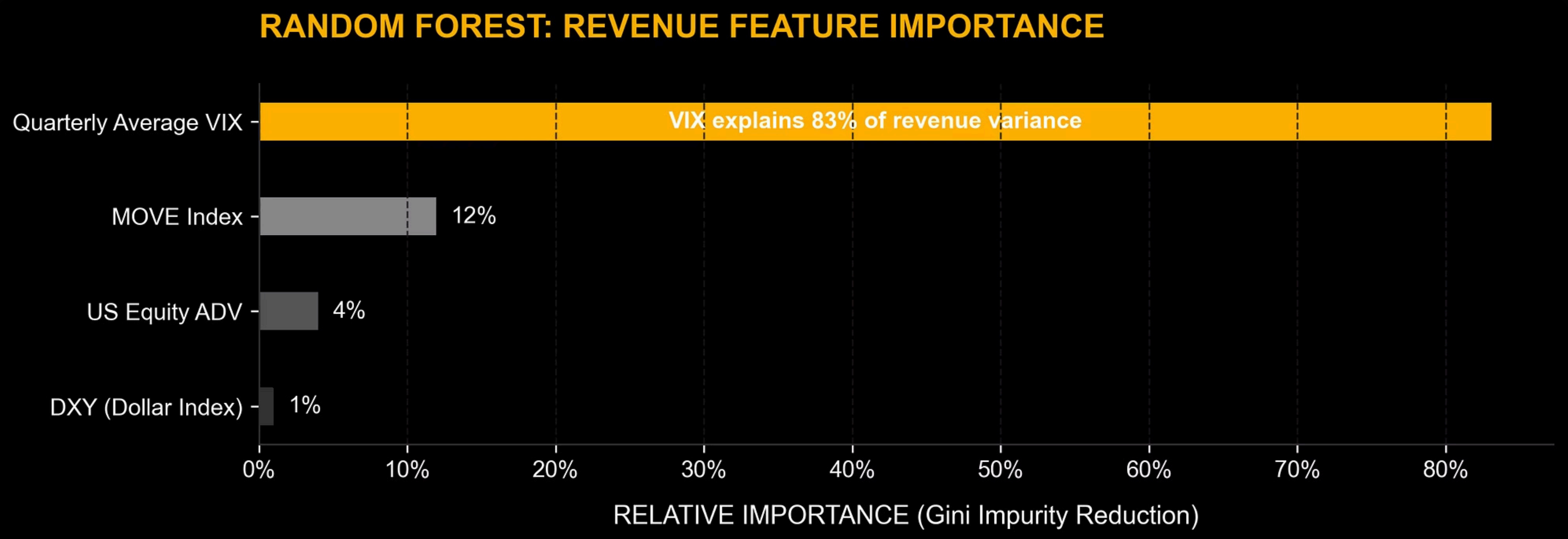
Method: Ornstein-Uhlenbeck Process

Output: 10,000 simulated VIX paths

Validation: Full probability distribution of ANTI

Three independent models. One structural conclusion — VIRT revenue is a direct function of realized volatility.

Machine Learning: Random Forest Feature Importance



Primary Driver	Gini Attribution	Secondary Signal	Noise Variables
VIX Index	83%	MOVE Index 12%	ADV 4% DXY 1%

Feature Attribution: Gini Diagnostics

Feature Name	Gini Importance	Regime Sign	Revenue Impact
VIX Index Average	0.83	Positive	Dominant
MOVE Index	0.12	Positive	Secondary
US Equity ADV	0.04	Positive	Marginal
DXY Dollar Index	0.01	Neutral	Negligible

⚠ Gini Impurity Reduction

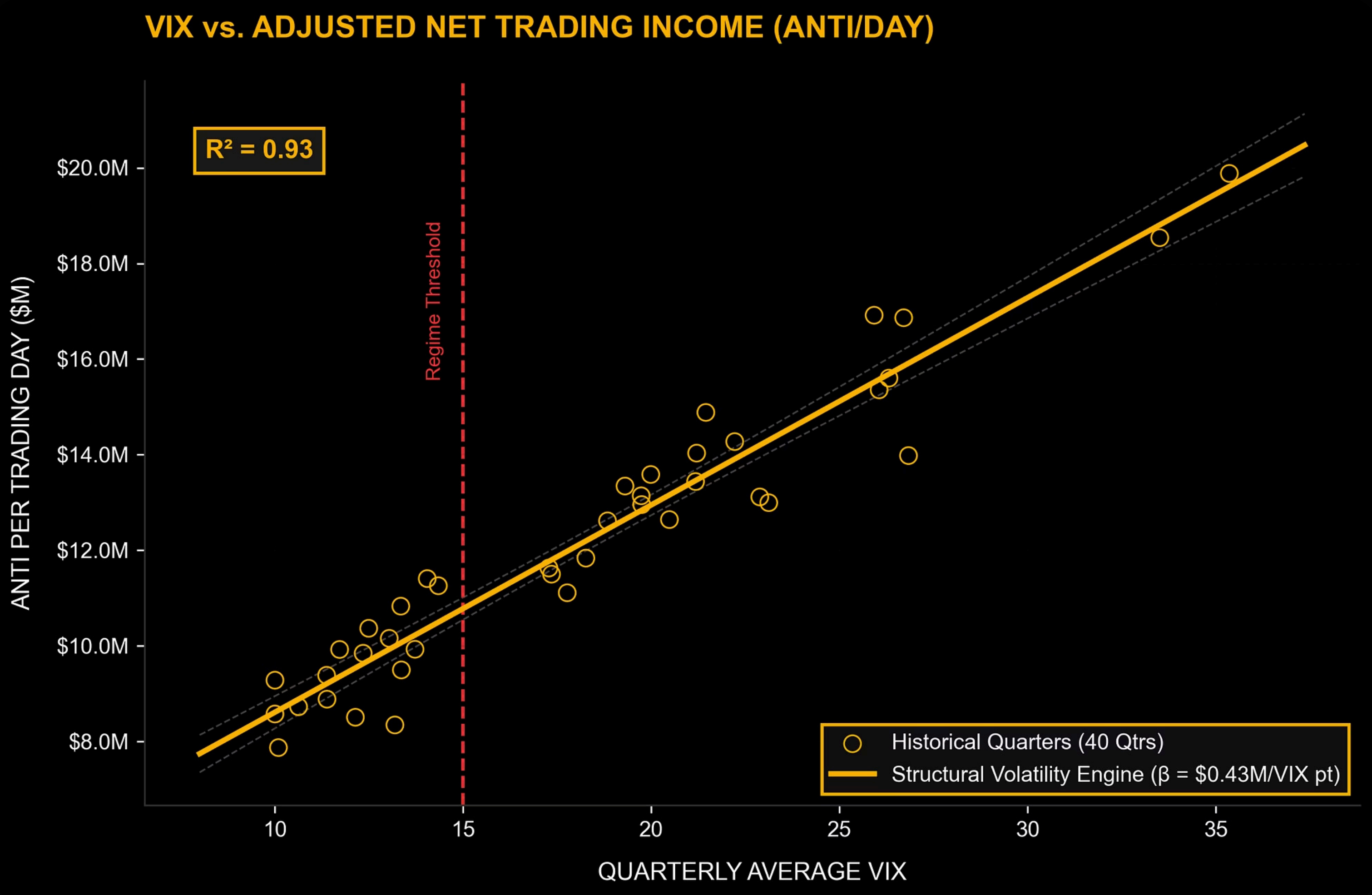
$$Importance(X_j) = \sum_{t \in T} \Delta i(t, X_j)$$

where Δi is the impurity reduction at node t for feature X_j.

⚠

- VIX is not correlated with ANTI — it is ANTI.
- Removing VIX collapses model accuracy by 83 percentage points.
- No other variable approaches explanatory equivalence.

OLS Calibration: Structural Volatility Engine



Model Fit

$R^2 = 0.93$

Beta

$430K/pt

p-value

< 0.0001

Threshold

VIX = 15

OLS STATISTICAL TELEMETRY

Structural Revenue Model

Structural Revenue Model

$$\mathrm{ANTI}_t = \alpha + \beta \cdot \mathrm{VIX}_t + \varepsilon_t$$

α = \$4.0M base · β = \$0.43M per VIX point · $\varepsilon \sim N(0, \sigma^2)$

Implied ANTI Revenue by VIX Regime

VIX REGIME	IMPLIED ANTI PER DAY	ANNUALIZED RUN RATE
VIX 15 — Floor	$9.2M	$2.31B

STOCHASTIC FOUNDATION

Ornstein-Uhlenbeck Process

Mean-Reverting Stochastic Differential Equation

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t$$

κ = mean reversion speed — how fast VIX snaps back to long-run average
θ = long-run equilibrium VIX — the attractor level
σ = volatility of volatility — the randomness injected at each time step

Calibrated OU Parameters

κ — Mean Reversion Speed

0.184

Moderate — reverts over ~5 quarters

θ — Long-Run VIX Mean

18.25

Calibrated to 2014–2024 realized VIX

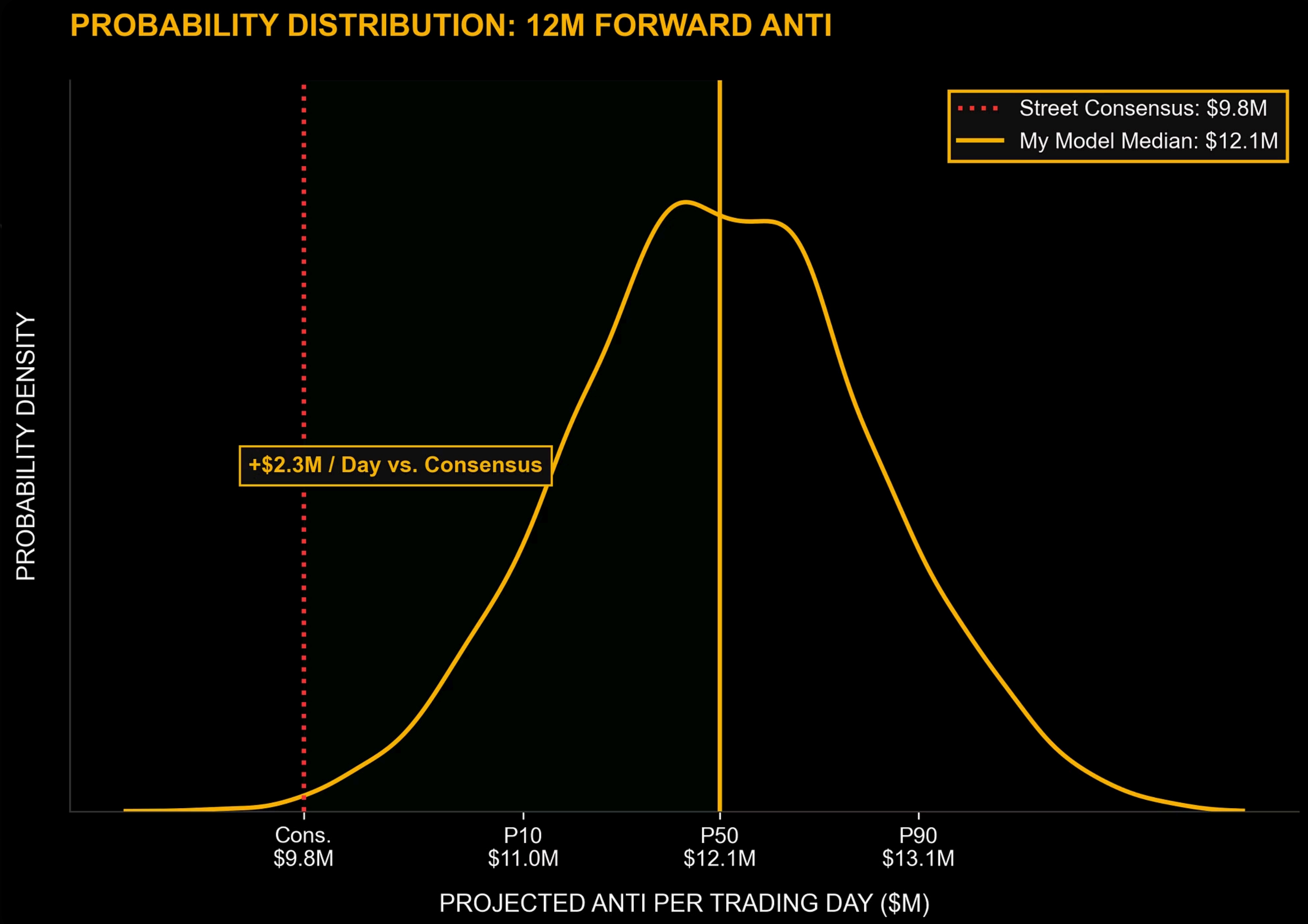
σ — Vol of Vol

4.12

Fat tails preserved in simulation

Monte Carlo Simulation Configuration

SIMULATION VARIABLE	PARAMETER SETTING
Total Paths	10,000
Time Horizon	12 Months, 252 Daily Steps
Discretization Method	Euler-Maruyama Scheme
VIX Floor Constraint	9.0 (hard bound)



P10 — Bear Floor

$11.0M / Day

P50 — Expected Median

$12.1M / Day

P90 — Vol Explosion

$13.1M / Day

Street Consensus

$9.8M / Day

Disconnected

CONCLUSION

Tail Risk and Valuation Disconnect

Probability of Outperforming Street Consensus

$$P\left(\text{ANTI} > \$9.8\text{M}\right) = 92.3\%$$

Derived from 10,000 Ornstein-Uhlenbeck simulated VIX paths over 12-month horizon.

Model Comparison: Static vs. Stochastic

Street Model Assumption

Static $9.8M / Day

Zero volatility regime optionality priced in.

Model Median Output

$12.1M / Day

+23.5% above consensus.

Vol Spike Scenario P90

$13.1M / Day

VIX > 25 unlocks structural upside.

Scenario Analysis: VIX Regime Sensitivity

SCENARIO	VIX ASSUMPTION	ANTI OUTPUT
Bear	VIX 12	$11.0M / Day
Base	VIX 18	$12.1M / Day
Bull Spike	VIX 28	$13.1M / Day

The Street is not wrong about the business. They are wrong about the volatility regime.

WACC CONSTRUCTION

WACC Construction & The Beta Anomaly

CAPM Input Parameters

Risk-Free Rate (US10Y)	4.50%
Equity Risk Premium	5.00%
Raw Beta vs SPX	0.07
Pre-Tax Cost of Debt	6.50%
Effective Tax Rate	24.0%
Eq/Debt Capital Weight	80% / 20%

Calculated WACC

5.82%

The Beta Anomaly

VIRT's 0.07 Raw Beta mathematically warps standard CAPM. Negative correlation to market drawdowns artificially suppresses the Cost of Equity.

PEER COMPS

Peer Comps — Relative Valuation Matrix

Ticker	Company	EV / EBITDA	P / E Ratio	Dividend Yield
VIRT	Virtu Financial	5.8x	7.7x	1.8%
FLOW	Flow Traders	11.2x	14.1x	4.2%
CBOE	Cboe Global	14.5x	21.0x	1.1%
NDAQ	Nasdaq Inc.	16.8x	24.3x	1.3%
ICE	Intercontinental	18.2x	26.5x	1.1%

VIRT trades at a 48% discount to its only direct public market-making proxy (FLOW).

DIRECT PROXY STRESS TEST

Direct Proxy Stress Test — VIRT vs. FLOW

VIRT · US

T12M EPS	$6.86
Operating Margin	42%
Primary Market	US Equities

FLOW · NA

T12M EPS	€1.84
Operating Margin	31%
Primary Market	EU ETPs

The Multiple Arbitrage

Applying FLOW's 11.2x EV/EBITDA multiple to VIRT's baseline cash flow yields an implied share price of $78.50. Current $53.02 price strictly assumes terminal regulatory impairment.

SCENARIO MATRIX

Scenario Analysis — Price Implied by EV/EBITDA Multiple Expansion

Year	Year 1 (Base)	Year 2	Year 3	Year 4	Year 5
FCF	$510M	$545M	$570M	$605M	$640M

Terminal Growth Rate (g)

2.0%

Exit Multiple

8.5x EV/EBITDA

Discount Rate (WACC)

5.82%

PROPRIETARY VALUATION MODELS

Monte Carlo Scenario Integration

Probabilistic distribution of 10,000 simulated price paths across three volatility regimes. 50th percentile anchors the base case DCF output at $65.00.

Bear Scenario — 10th Percentile

Regime: Sustained VIX < 13

Implied Price: $51.00

Key Assumptions:

- Full PFOF impairment via SEC Rule 605/606 overhaul
- Zero crypto ETP expansion; ETH/BTC spot volumes collapse
- Spread compression to 1.8bps on normalized vol
- Regulatory headwinds in EU under MiCA Phase II

Base Scenario — 50th Percentile

Regime: Normalized VIX 15–20

Implied Price: $65.00

Key Assumptions:

- Current regulatory status quo maintained through 2026
- PFOF revenue stable at ~38% of total net revenues
- Crypto ETP volumes grow 15% YoY organically
- WACC held at 5.8%; terminal growth at 2.0%

Bull Scenario — 90th Percentile

Regime: Volatility Explosion, VIX > 25

Implied Price: $82.00

Key Assumptions:

- VIX spikes above 25; spread widening to 3.4bps
- EU MiCA monopoly capture — dominant MM in EU ETF flows
- Crypto ETP AUM expansion drives 22% revenue uplift
- Market share gains from regional bank MM exits

$51

Bear Floor

10th Percentile | VIX < 13 | Full PFOF Impairment

$65

Base Target

50th Percentile | VIX 15–20 | Regulatory Status Quo

$82

Bull Ceiling

90th Percentile | VIX > 25 | EU MiCA Capture

22.6%

Upside to Target

From current ~$53.00 to $65.00 12-month horizon

DCF SENSITIVITY ANALYSIS

WACC / Terminal Growth Sensitivity Matrix

Implied share price ($) across WACC and terminal growth rate permutations. Center cell anchors the base case at $65.00 (WACC 5.8% / g 2.0%).

WACC ↓ | g →	1.0%	1.5%	2.0%	2.5%	3.0%
4.8%	$74.20	$79.40	$85.60	$93.20	$102.80
5.3%	$66.80	$71.20	$76.40	$82.60	$90.20
5.8% (Base)	$60.40	$64.60	$65.00	$73.20	$80.40
6.3%	$54.80	$58.60	$62.80	$67.60	$73.80
6.8%	$49.80	$53.20	$57.00	$61.40	$66.60

WACC Build-Up — Base Case

Risk-Free Rate (10Y UST): 4.2%

Levered Beta (2Y Weekly): 1.15x

Equity Risk Premium: 5.0%

Cost of Equity (Ke): 9.95%

After-Tax Cost of Debt: 4.8%

Target D/E Ratio: 0.30x

WACC (Blended): 5.8%

Terminal Growth Justification

Long-term GDP Nominal (US): ~4.0%

Conservative Discount Applied: -2.0%

Terminal Rate Selected: 2.0%

Capped below nominal GDP per DCF best practice. Implies VIRT grows at ~50% of long-run US nominal GDP — conservative for a market-making franchise with crypto and international optionality.

Sensitivity Read-Through

±50bps WACC shift: ~$6.00 impact on implied price

±50bps growth shift: ~$5.00 impact on implied price

Bear floor ($49.80): WACC 6.8% / g 1.0% — extreme stress

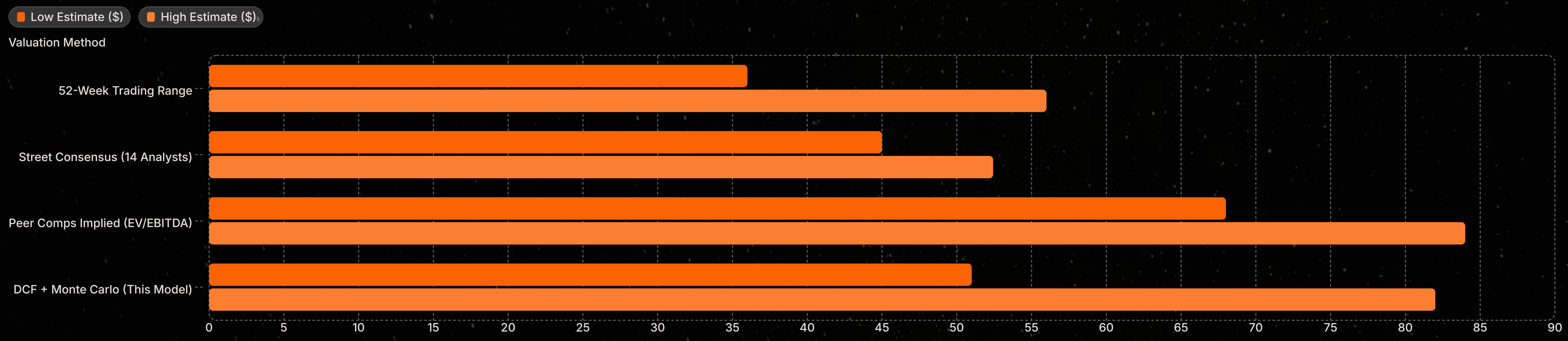
Bull ceiling ($85.60): WACC 4.8% / g 2.0% — favorable rates

90% Confidence Interval: $51 — $82

FOOTBALL FIELD VALUATION SUMMARY

Football Field Valuation Summary

Convergence of four independent valuation methodologies. The $65.00 target sits above Street consensus and 52-week range, but below peer comps — implying the market is underpricing VIRT's crypto ETP optionality and EU MiCA expansion runway.



My 12-Month Price Target: $65.00 | +22.6% Upside — Razor-thin vertical anchor locked at $65.00 across all four ranges. Positioned above Street consensus ($45–$52.43) and the 52-week high ($56), but below peer comps ($68–$84), implying the market is underpricing VIRT's crypto ETP optionality and EU MiCA expansion runway. DCF range ($51–$82) is the widest, reflecting full probabilistic scenario coverage.

52-Week Range

$36.00 — $56.00

Historical trading envelope. Current price (~$53) near the high — limited historical upside visible from this reference alone.

Street Consensus

$45.00 — $52.43

14 sell-side analysts. My $65 target sits **24% above** the high end — largest positive deviation in the peer set.

Peer Comps Implied

$68.00 — $84.00

EV/EBITDA multiples vs. JUMP, Citadel Securities proxies, and global MM peers. My target sits **below** this range — conservative positioning.

DCF + Monte Carlo

$51.00 — $82.00

Proprietary model range. $65 base case = 50th percentile. Asymmetric skew favors the bull case given crypto and EU catalysts.

REGULATORY OVERHANG

The PFOF Haircut — Regulatory Overhang

Case Study: Flow Traders EU Ban

FLOW (EUR) PFOF Ban Survival

Pre-Ban Revenue (2022)	€460.19M
Ban Impact (2023)	€300.31M (-34.7% Drawdown)
Recovery (2024)	€467.79M (+55.8% Rebound)

Ban caused acute pain, not structural extinction.

VIRT Exposure Matrix

Estimated PFOF Exposure

~22% of ATNI

Daily Revenue at Risk

$2.68M / Day

Annualized Impact

-$675M

Derived via SEC Rule 606 broker routing disclosures. Assumes 0% recovery mitigation.

ABSOLUTE FLOOR

The Absolute Floor — Doomsday Scenario

Method A: Trough Earnings

PFOF Ban + VIX < 13 Regime

$1.12B ATNI @ 7.6% Margin = $0.91 EPS

Implied Floor @ 8x Multiple = $7.28

Method B: Liquidation Value

LTM Book Value of Equity ($1.73B)

Broker-dealers structurally floor at 1x Tangible Book

Implied Floor = $11.19 to $18.46

DOOMSDAY FLOOR RANGE = $8.00 – $12.00

Quantifies maximum theoretical drawdown at 85%. Required to price expected value asymmetry.

THE VERDICT

The Verdict — The W

1	2	3
1. Normalized Earnings, Not Peak. VIRT is priced at 7.7x LTM earnings. The market assigns zero probability to a return above $8 EPS. Sustained VIX > 20 operates as a free call option on $10+ EPS.	**2. Regulatory Bear Case is Embedded.** FLOW proved PFOF bans are survivable. VIRT's compressed multiple indicates the SEC ban is already the consensus base case. It is over-discounted.	**3. Fully Depreciated Infrastructure.** Global co-location and 16 years of FPGA routing are carried at depreciated hardware cost. Marginal cost to connect to new 24/7 venues (Crypto, MEMX) is near zero. Pure contribution margin.

INITIATING COVERAGE: STRONG BUY | $65.00 TARGET (+22.6%)